07
PHASE

Storytelling, Communication & Domain Knowledge

Turn Insights Into Impact.
Communicate. Influence. Inspire.

- Tell Compelling Stories
- Communicate Clearly
- Drive Better Decisions
- Bring Domain Knowledge to Life

- Simplify Complex Data
- Visualize Effectively
- Communicate with Impact
- Influence Stakeholders
- Apply Domain Knowledge

Great Data Tells a Story. Great Analysts Drive Change.

U N I T 1

Problem Framing & The Art of the Question

Topics Covered	Sections per Topic
5 Core Topics	**7 Structured Sections**

Unit 1 | Topic 1.1

The "5 Whys" Technique

Digging past the surface request to find the true business pain.

1 | Definition

The 5 Whys technique is a simple but powerful problem-solving method where you repeatedly ask "Why?" (typically five times) to move from a surface-level symptom to the actual root cause of a problem.

It was originally developed by Sakichi Toyoda and used within Toyota's manufacturing system. For data analysts, it acts as a mental drill — helping you understand what a business stakeholder really needs before you touch any data.

2 | Sub-Topics

- **Iterative Questioning: Each answer becomes the input to the next "Why?" — you are building a chain of cause-and-effect.**
- Root Cause vs. Symptom: A symptom is what you observe (e.g., revenue dropped). The root cause is WHY that happened (e.g., a pricing error in one product category).
- Stopping Criteria: You stop asking when you reach an actionable cause — something the team can actually fix or investigate.
- Fishbone (Ishikawa) Diagram: A visual companion to the 5 Whys, often used together to map multiple causes branching from one problem.
- 5 Whys in Agile/Data Context: Used in sprint retrospectives and analysis kick-offs to ensure the right question is being answered.

3 | Why Important for a Data Analyst

Without this technique, analysts risk solving the wrong problem — building charts and dashboards that look great but don't help the business. The 5 Whys ensures your analysis has real business impact.

Business Risk (Without 5 Whys)	Benefit (With 5 Whys)
Answer the wrong question	Solve the actual business problem

Waste days on irrelevant analysis	Save time by focusing correctly
Deliver insights no one acts on	Deliver actionable, trusted insights
Lose stakeholder trust	Build credibility as a strategic analyst
Create confusion about priorities	Align team on what truly matters

4 | Real-World Analogy

> Think of a doctor. When you say "I have a headache," the doctor does not immediately prescribe a painkiller. They ask: Is it a tension headache? Are you dehydrated? Do you have high blood pressure? Is it stress-related? Each question digs deeper. The doctor treats the root cause — not just the headache. A data analyst does the same thing with business problems.

5 | Framework / Syntax

The 5 Whys follows this structured chain pattern:

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PROBLEM STATEMENT: [Describe the observed symptom clearly]

Why 1: Why is [symptom] happening?
 → Answer 1: [First-level cause]

Why 2: Why is [Answer 1] happening?
 → Answer 2: [Deeper cause]

Why 3: Why is [Answer 2] happening?
 → Answer 3: [Even deeper cause]

Why 4: Why is [Answer 3] happening?
 → Answer 4: [Near root cause]

Why 5: Why is [Answer 4] happening?
 → Root Cause: [Actionable root cause - STOP HERE]

ACTION: [What data question do we now ask?]
```

6 | Practical Example

Scenario: The VP of Sales says — "Our revenue dropped last quarter. Can you pull me a report?"

Why?	Answer
Why did revenue drop?	Fewer customers completed purchases
Why did fewer customers complete purchases?	Checkout abandonment rate increased by 35%
Why did checkout abandonment increase?	Users were dropping off at the payment step
Why were they dropping off at payment?	A new payment gateway was introduced last quarter
Why did the new gateway cause drop-offs?	It added 3 extra form fields and was slow on mobile

Root Cause: The new payment gateway added friction on mobile devices. NEW DATA QUESTION: "What is the mobile vs. desktop checkout completion rate before and after the gateway change, broken down by device type and payment step?" — This is now a focused, actionable analysis.

7 | Industry Use Cases

Industry	Application of 5 Whys
E-Commerce / Retail	Why are product returns rising? → Root cause: product descriptions mismatch actual items → Fix: audit listing content quality
Banking / Finance	Why is loan approval declining? → Root cause: credit score model not updated for post-pandemic income patterns
Healthcare	Why are patient readmissions high? → Root cause: discharge instructions not understood → Fix: analyze literacy levels vs. instructions
Manufacturing	Why are defects increasing? → Root cause: raw material supplier changed formulation → Fix: quality control data by batch
SaaS / Technology	Why is churn increasing? → Root cause: users cannot complete onboarding → Fix: analyze drop-off by onboarding step

Marketing / Advertising	Why did campaign ROI fall? → Root cause: audience segment over-saturated → Fix: frequency capping analysis

Topic 1.2

Converting Business Problems into Data Questions

From vague business pain to precise, answerable data questions.

1 | Definition

Converting business problems into data questions is the skill of taking a vague, high-level business concern — like "Why are we losing money?" — and rephrasing it as a specific, measurable, and answerable question that can be explored using data — like "Which customer segment has the highest churn rate this quarter?"

This translation step is critical. Business stakeholders speak in outcomes and feelings. Data analysts need to speak in metrics, dimensions, and filters. Bridging this gap is what separates a good analyst from a great one.

2 | Sub-Topics

- **Business Language vs. Data Language: Business people use words like "performance," "growth," "problems." Analysts reframe these into measurable quantities.**
- The SMART Question Framework: A good data question is Specific, Measurable, Achievable (with available data), Relevant (to the problem), and Time-bound.
- Decomposing Vague Problems: Breaking a large fuzzy problem into 2-4 specific sub-questions that together answer the full problem.
- Identifying the Right Metric: Understanding which number best represents business health (e.g., using Churn Rate instead of Total Revenue to understand customer loss).
- Dimensioning the Question: Adding who, what, when, and where to a question to make it precise (e.g., "Which region?" "Which age group?" "In Q3?").

3 | Why Important for a Data Analyst

If you analyze data without a precise question, you produce reports that describe the data instead of answering the problem. Stakeholders will say "interesting" and do nothing with it. A precise data question gives your analysis direction, purpose, and actionability.

4 | Real-World Analogy

> Imagine a chef saying "The food doesn't taste right." That's a business problem. The kitchen analyst (sous-chef) must turn this into a specific question: "Is the salt level incorrect in the soup specifically, based on today's batch?" Only then can the cook fix the exact issue. Without the specific question, they might change everything and ruin other dishes.

5 | Conversion Framework / Syntax

Use this translation template to convert any business problem into a data question:

```
BUSINESS PROBLEM: [Vague concern from stakeholder]

STEP 1 — Identify the Business Outcome
 What outcome is the business worried about?
 e.g., "We are losing customers"

STEP 2 — Identify the Key Metric
 What number measures this outcome?
 e.g., Churn Rate = (Customers Lost / Total Customers) x 100

STEP 3 — Add Dimensions (Who / What / When / Where)
 Which segment? Which time period? Which product?
 e.g., By customer segment + by quarter

STEP 4 — Frame the Data Question
 "[Metric] for [Dimension] during [Time Period]?"
 e.g., "Which customer segment has the highest churn rate in Q3?"

STEP 5 — Confirm Answerability
 Do we have the data to answer this? → Yes / No / Partial
```

6 | Practical Example

Business Problem	Data Question
Why are we losing money?	Which product category has the highest return rate and lowest margin in the last 6 months?
Our app is not performing well	What is the average session duration and DAU/MAU ratio by platform (iOS vs Android) this quarter?
Customers seem unhappy	Which customer segment has an NPS below 30, and what feature do they use most before churning?
Marketing is not working	Which campaign source has a cost-per-acquisition above the industry average of $45 this month?
We are growing but not fast enough	What is the month-over-month growth rate of new paid users in the SMB segment vs Enterprise?

7 | Industry Use Cases

Industry	Business Problem → Data Question
Retail / E-Commerce	"Sales are slow" → "Which product subcategory has below-average conversion rate during weekday evenings?"
Healthcare	"Patient outcomes are poor" → "What is the 30-day readmission rate for patients over 65 in the cardiology ward?"
EdTech / Education	"Students are dropping out" → "At which course module do learners have the highest dropout rate by subject?"
Logistics	"Deliveries are late" → "Which delivery routes have >15% late delivery rate when temperature is above 35°C?"
HR / People Analytics	"Employee morale is low" → "What is the resignation rate by department and tenure band in the last 12 months?"

Topic 1.3

Defining Success Metrics: Setting Clear KPIs

Setting clear KPIs before the analysis begins — so everyone knows what "good" looks like.

1 | Definition

A Success Metric (also called a KPI — Key Performance Indicator) is a specific, measurable number that tells you whether a goal has been achieved. Before starting any analysis, you must agree with stakeholders on what success looks like — otherwise, you won't know when the analysis is "done" or whether the results are good or bad.

KPIs turn fuzzy goals like "grow the business" into precise targets like "increase monthly active users by 15% in Q4."

2 | Sub-Topics

- **Metrics vs. KPIs: A metric is any measurable number (e.g., page views). A KPI is a metric tied to a strategic business goal with a target (e.g., page views must reach 1M/month by Q4).**
- Leading vs. Lagging Indicators: Lagging = measures past performance (Revenue last quarter). Leading = predicts future performance (Number of proposals sent this week).
- North Star Metric: The single most important metric for a company or product (e.g., Airbnb's North Star = Nights Booked; Facebook's = Daily Active Users).
- Guardrail Metrics: Secondary metrics that must NOT decline even while optimizing for the primary goal (e.g., you improve conversion rate, but don't let customer satisfaction fall).
- SMART KPIs: Specific, Measurable, Achievable, Relevant, Time-bound — the gold standard for KPI definition.
- Metric Hierarchies: Business metrics cascade from company-level (Total Revenue) → Team-level (Regional Sales) → Individual-level (Sales per Rep).

3 | Why Important for a Data Analyst

If you start an analysis without defined KPIs, you risk "analysis drift" — endlessly exploring data without reaching a conclusion. Defined KPIs give your analysis a finish line. They also prevent stakeholders from moving the goalposts after you deliver results.

> ⚠️ A common analyst mistake: delivering a detailed report and then being told "But we wanted to measure something else." Clear KPIs agreed upon upfront prevent this entirely.

4 | Real-World Analogy

Think of running a race. Without a finish line, how do you know when you've won? KPIs are the finish line. "Run faster" is vague. "Run 10km in under 50 minutes by December" is a KPI — Specific (10km), Measurable (time), Achievable, Relevant (fitness goal), Time-bound (by December). Now you know exactly what to train for.

5 | KPI Definition Framework

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KPI DEFINITION TEMPLATE
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KPI Name : [Short, clear name e.g., "Customer Churn Rate"]

Definition : [How is this metric calculated?]
 Formula: (Customers Lost in Period / Total Customers at Start) x 100

Owner : [Who is responsible for this metric? e.g., Head of CX]

Data Source : [Where does the data come from? e.g., CRM Database]

Frequency : [How often is it measured? Daily / Weekly / Monthly]

Current Value : [Baseline - what is it today? e.g., 8.5%]

Target Value : [What should it be? e.g., Below 5%]

Time Bound : [By when? e.g., End of Q4 2025]

Guardrail : [What metric must NOT decline? e.g., NPS must stay above 40]
```

6 | Practical Example

Scenario: An e-commerce company wants to "improve customer loyalty."

KPI Component	Value
KPI Name	Customer Retention Rate
Formula	((Customers at End - New Customers) / Customers at Start) × 100
Owner	Head of Customer Success
Data Source	Order Management System + CRM
Frequency	Monthly
Current Value (Baseline)	72%
Target Value	80% or above
Time Bound	By end of Q4 2025
Guardrail Metric	Average Order Value must not drop below $85

7 | Industry Use Cases

Industry	Key KPIs Used
SaaS / Software	MRR (Monthly Recurring Revenue), Churn Rate, Customer Acquisition Cost (CAC), LTV:CAC Ratio
E-Commerce / Retail	Conversion Rate, Cart Abandonment Rate, Average Order Value (AOV), Return Rate
Banking / Finance	Non-Performing Loan Ratio, Customer Lifetime Value, Net Interest Margin, Fraud Detection Rate
Healthcare	30-Day Readmission Rate, Bed Occupancy Rate, Average Length of Stay, Patient Satisfaction Score
Marketing	Cost Per Acquisition (CPA), Click-Through Rate (CTR), Return on Ad Spend (ROAS), Lead Conversion Rate
HR / People Analytics	Employee Turnover Rate, Time-to-Hire, Engagement Score, Absenteeism Rate

Topic 1.4

Stakeholder Interviewing: Running a Discovery Session

How to ask the right questions to the right people before any analysis begins.

1 | Definition

Stakeholder Interviewing (also called a Discovery Session) is a structured conversation held with business stakeholders — managers, executives, product owners, or domain experts — before an analysis begins. The goal is to understand their real needs, expectations, constraints, and definition of success.

A stakeholder is anyone who has an interest in the analysis or will be affected by its outcome. They are often non-technical and may not know how to phrase what they need in data terms — your job is to translate their world into data questions.

2 | Sub-Topics

- **Types of Stakeholders: Primary (decision-makers who will act on findings), Secondary (those affected by decisions), Technical (engineers/IT who own the data).**
- Open vs. Closed Questions: Open questions ("Tell me more about the problem") uncover context. Closed questions ("Is this a monthly or quarterly metric?") confirm specifics.
- The Discovery Session Agenda: A structured meeting with a clear purpose, order of questions, and agreed outcomes (see framework below).
- Active Listening: Paraphrasing, summarizing, and reflecting what the stakeholder says to ensure you've understood correctly.
- Managing Scope Creep: Stakeholders often keep adding new questions during discovery. You need techniques to acknowledge requests and defer them to a future phase.
- Documentation: Writing a Discovery Summary and sharing it back with stakeholders for sign-off before analysis starts.

3 | Why Important for a Data Analyst

Without a proper discovery session, analysts make assumptions. Assumptions lead to misaligned analyses. You might spend two weeks building a beautiful dashboard only to be told "That's not what I meant." Stakeholder interviews prevent this.

Problem Without Discovery	Solution With Discovery
Analyst assumes the question	Question is co-created with stakeholder
Wrong metric is analyzed	Right metric agreed upon upfront
Output is irrelevant	Output directly answers business need
Rework wastes time	First delivery is on target
Analyst loses credibility	Analyst is seen as strategic partner

4 | Real-World Analogy

> A good architect does not start drawing a house after the client says "Build me a nice house." They sit down and ask: How many people live here? Do you need a home office? What's your budget? Do you prefer open-plan or separate rooms? Only then do they put pencil to paper. A data analyst's discovery session is the architect's client brief — it defines everything that follows.

5 | Discovery Session Framework

Use this structured agenda for a 45–60 minute discovery session:

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DISCOVERY SESSION AGENDA (45–60 minutes)
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[ 0–5 min ] OPENING
 "Thanks for your time. The goal today is for me to fully understand
 your needs so the analysis hits the mark. May I take handbook?"

[ 5–15 min ] CONTEXT GATHERING (Open Questions)
 Q1: "In your own words, what problem are you trying to solve?"
 Q2: "How is this problem currently impacting the business?"
 Q3: "What have you already tried to address this?"

[ 15–30 min ] DEEP DIVE (Probing Questions)
 Q4: "Who are the end users of this analysis?"
 Q5: "What decision will be made based on these findings?"
 Q6: "What does a successful outcome look like to you?"
 Q7: "Are there any metrics you already track for this?"

[ 30–40 min ] CONSTRAINTS CHECK
 Q8: "Are there any data sources you know should be included?"
 Q9: "When do you need this by?"
 Q10: "Are there topics we should NOT include?"

[ 40–50 min ] SUMMARY + VALIDATION
```

```
 Paraphrase back: "So if I understood correctly, you need [X]
 by [date], and success means [Y]. Is that right?"

[ 50-60 min ] NEXT STEPS
 "I will send a written summary by [date] for your sign-off."
```

6 | Practical Example

Scenario: Marketing Director says "Our email campaigns aren't working."

Question Asked	Stakeholder Answer	Analyst Takeaway
What does "not working" mean to you?	Open rates are low and nobody buys	Metrics: Open Rate, Conversion Rate
What campaigns are we talking about?	Mainly the weekly product launch emails	Scope: Weekly product emails only
Who are these emails sent to?	All subscribers, about 200,000	Segment: Full subscriber list
What would success look like?	Open rate above 25%, 3x more purchases	Target KPIs defined clearly
When do you need this?	Before next campaign — 2 weeks away	Timeline confirmed: 2 weeks

7 | Industry Use Cases

Industry	Discovery Session Application
Consulting / Advisory	Run discovery with client C-suite before proposing any analytics solutions or dashboards
Product / SaaS	Interview Product Manager before building feature adoption analysis to understand feature prioritization goals
Retail / FMCG	Interview Category Manager to understand promotional pricing objectives before building promotional lift model
Healthcare Administration	Interview clinical directors before analyzing patient flow data to understand care pathway priorities
Financial Services	Interview Risk Manager before building fraud detection model to understand acceptable false positive rates

Topic 1.5

Identifying Constraints

Understanding data availability, time sensitivity, and budget before analysis begins.

1 | Definition

Constraints are real-world limitations that define the boundaries within which a data analysis must be completed. Every analysis project faces three core types of constraints: Data Availability (what data exists and can be accessed), Time Sensitivity (when the results are needed), and Budget (resources available — tools, people, compute).

Identifying constraints early is crucial because they determine the scope, depth, and method of your analysis. Ignoring them leads to over-promising and under-delivering — one of the most damaging mistakes an analyst can make.

2 | Sub-Topics

- **Data Availability Constraints: Does the required data exist? Is it stored? Is it clean? Do you have access permissions? How far back does history go?**
- Time Constraints: Hard deadline (non-negotiable date), Soft deadline (preferred date), and Time-to-Insight (how long the analysis will realistically take).
- Budget Constraints: Tool costs (SQL licenses, BI tools), Compute costs (cloud processing for large datasets), and Analyst hours available.
- Technical Constraints: Data formats, database access, API rate limits, privacy/security restrictions (e.g., GDPR, HIPAA).
- Organizational Constraints: Stakeholder approval chains, legal/compliance review requirements, or dependency on another team's data pipeline.
- The Constraint Triangle: A classic project management model — Scope, Time, and Cost. You can optimize two but must sacrifice one. Analysts face this constantly.

3 | Why Important for a Data Analyst

Analysts who identify constraints early can set realistic expectations, propose alternative approaches if data is missing, and avoid the trap of promising a full analysis in an impossible timeframe. It protects both the analyst and the business from disappointment.

> ⚠ Real scenario: An analyst promises a 3-year trend analysis. Halfway through, they discover the database only holds 18 months of data. The entire approach must change. Identifying data availability upfront avoids this. Always audit your data before committing to a scope.

4 | Real-World Analogy

> Imagine a chef hired to cook a 5-course dinner for 50 guests by 7pm with a budget of ₹10,000. The constraints are: ingredients available in the kitchen (data availability), dinner served by 7pm (time sensitivity), and ₹10,000 total (budget). If the chef doesn't check the pantry first, they may plan a dish requiring an ingredient that isn't there. A smart chef identifies constraints first, then designs the best possible menu within them.

5 | Constraint Assessment Framework

Use this checklist at the start of every analysis project:

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CONSTRAINT ASSESSMENT CHECKLIST
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A. DATA AVAILABILITY
 [ ] What data is needed for this analysis?
 [ ] Where does this data live? (Database / File / API / Manual)
 [ ] Do I have access? (Request access if not)
 [ ] How far back does the history go?
 [ ] What is the data quality? (Missing values? Duplicates?)
 [ ] Are there privacy/compliance restrictions on this data?

B. TIME CONSTRAINTS
 [ ] What is the hard deadline?
 [ ] What is the preferred delivery date?
 [ ] How long will data extraction take?
 [ ] How long will cleaning and analysis take?
```

```
  [ ] Is the timeline feasible? If not → scope down or get more resources.

C. BUDGET / RESOURCE CONSTRAINTS
  [ ] What tools are available? (SQL, Python, Tableau, etc.)
  [ ] How many analyst hours are allocated?
  [ ] Are there costs for cloud compute or licensed tools?
  [ ] Are other team members available to help?

D. ORGANIZATIONAL CONSTRAINTS
  [ ] Are there approvals needed before accessing certain data?
  [ ] Does legal/compliance need to review the approach?
  [ ] Is there a dependency on another team's deliverable?

OUTCOME: Document all constraints → Share with stakeholder → Adjust scope.
```

6 | Practical Example

Scenario: Stakeholder requests a 5-year customer retention trend analysis to be delivered in 3 days.

Constraint Type	Finding	Impact & Resolution
Data Availability	CRM only stores 2 years of customer records	Scope reduced to 2-year trend. Stakeholder informed and approves.
Data Quality	18% of customer records have missing join dates	Analyst flags data quality issue. Filters to clean records only and handbook caveat in report.
Time Constraint	3-day deadline for 5-year analysis is unrealistic	Negotiate: deliver preliminary 2-year insight in 3 days, full analysis in 10 days.
Access Constraint	Customer PII is restricted under GDPR policy	Work with anonymized Customer IDs only. Request data from data governance team.
Budget Constraint	Large dataset requires cloud compute — not budgeted	Use sampling (random 20% sample) for initial analysis to validate approach first.

7 | Industry Use Cases

Industry	Constraint Challenge & Resolution
Banking / Finance	GDPR and RBI data privacy rules restrict customer data use. Analysts work with anonymized, aggregated datasets and pre-approved query patterns.
Healthcare	HIPAA regulations in the US require patient data to be de-identified. Analysis is run on tokenized patient IDs only.
Retail / FMCG	Historical POS data only retained for 13 months. Analysts note this in any trend report and flag when year-over-year comparisons are affected.
Startups	No historical data exists (new company). Analysts design data collection strategy first and use industry benchmarks as proxy baselines.
Government / Public Sector	Data access requires multi-level approvals. Analysts build in 2-week approval lead time into every project plan.
Logistics	Real-time GPS data is too large to process in standard tools. Analysts sample by route and time window to make analysis computationally feasible.

UNIT 1 — Problem Framing & The Art of the Question

Topics: 5 Whys | Business→Data Questions | KPIs | Stakeholder Interviews | Constraints

Total Questions: 20 | Format: MCQ + Descriptive + Coding + DB

LEVEL 1: Warmup Test

5 Multiple Choice Questions — Foundational Understanding

Q1 What is the primary purpose of the '5 Whys' technique in data analysis?

A) To create visual dashboards quickly
B) To move from a surface-level symptom to the actual root cause of a problem
C) To define budget constraints before analysis
D) To identify the number of stakeholders in a project

ANS

Q2 Which framework is used to ensure a data question is well-formed and answerable?

A) GDPR Framework
B) Fishbone Diagram
C) SMART Question Framework
D) Agile Sprint Retrospective

ANS

Q3 What is the difference between a 'Metric' and a 'KPI'?

A) Metrics are visualizations; KPIs are raw numbers
B) A metric is any measurable number; a KPI is a metric tied to a strategic goal with a target
C) KPIs are used only in healthcare; metrics are used everywhere
D) Both terms are interchangeable and mean the same thing

ANS

Q4 In the Constraint Triangle model for analysis projects, which of the following is correct?

A) You can fully optimize Scope, Time, and Cost simultaneously
B) Only Time and Budget matter; Scope is flexible
C) You can optimize two but must sacrifice one
D) The triangle applies only to software engineering, not data analysis

ANS

Q5 What is a 'Guardrail Metric' in the context of KPI setting?

A) The primary metric the entire company optimizes for
B) A metric used only in manufacturing quality control
C) A secondary metric that must NOT decline while optimizing the primary goal
D) A leading indicator that predicts future performance

ANS

LEVEL 2: Interviewer Test

15 Intermediate to Semi-Advanced Questions — Descriptive, Scenario, Coding & Database

SECTION A — Descriptive Questions (4 Questions)

Q1 [Descriptive] Explain the '5 Whys' technique and how a data analyst should apply it before starting an analysis. Why is stopping at the first 'Why' dangerous?

Q2 [Descriptive] What is the difference between a 'Leading Indicator' and a 'Lagging Indicator'? Give an industry example of each and explain when an analyst should prioritize one over the other.

ANS

Q3 [Descriptive] Describe the complete process of converting a vague business problem into a precise, answerable data question. Walk through each step using the SMART criteria and dimensioning approach.

ANS

Q4 [Descriptive] Why is a Stakeholder Discovery Session important before starting any data analysis? What are the risks of skipping it, and what are the key questions an analyst must ask during such a session?

ANS

SECTION B — Scenario-Based Descriptive Questions (4 Questions)

Q1 [Scenario] Scenario: The Head of Marketing at an e-commerce company tells you: 'Our email campaigns aren't working. Can you look into it?' Walk through how you would apply the 5 Whys technique and then convert the finding into a precise data question.

ANS

Q2 [Scenario] Scenario: You are a new data analyst at a healthcare startup. The Clinical Director says: 'We need to reduce patient readmissions. Can you run an analysis?' Identify the constraints you would check before starting, and explain how you would present these constraints to the Clinical Director.

ANS

Q3 [Scenario] Scenario: A Product Manager tells you: 'Users are churning. Our retention is terrible.' Define three KPIs you would propose for this situation. For each KPI, specify the formula, target, and a guardrail metric.

ANS

Q4 [Scenario] Scenario: The VP of Sales requests a '5-year revenue trend analysis' to be delivered in 3 business days. After checking your data systems, you discover the database only holds 2 years of data, and 15% of revenue records have a missing region field. How do you handle this professionally?

ANS

SECTION C — Coding Questions (4 Questions)

Q1 [Coding] Write a Python function that implements the 5 Whys chain interactively. The function should accept a problem statement, prompt the user to enter answers for each 'Why', and return the root cause along with the full chain.

ANS

Q2 [Coding] Write a Python function that takes a business problem description as input and generates a SMART data question by prompting the user for metric, dimension, and time period. Validate that all SMART criteria are met before returning.

ANS

Q3 [Coding] Write a Python class called 'KPITracker' that allows analysts to define KPIs, log current values, check if targets are met, and flag guardrail breaches. Include methods: add_kpi(), log_value(), check_status(), and summary_report().

ANS

Q4 [Coding] Write a Python function 'constraint_audit()' that accepts analysis parameters (required_history_years, deadline_days, records_count) and evaluates feasibility. It should return a structured constraint report with recommendations.

ANS

SECTION D — Database Scenario-Based Questions (3 Questions)

Q1 [DB Coding] Database Scenario: You have a table 'orders' with columns: order_id, customer_id, order_date, total_amount, product_category, region, status. Write a SQL query to identify the top 3 product categories by return rate (status = 'Returned') for the last 6 months, along with total returns and total orders per category.

ANS

Q2 [DB Coding] Database Scenario: You have two tables — 'customers' (customer_id, signup_date, plan_type, region) and 'subscriptions' (sub_id, customer_id, start_date, end_date, status). Write a SQL query to calculate the monthly churn rate for the last 12 months, broken down by plan_type. Define 'churned' as status = 'cancelled'.

ANS

Q3 [DB Coding] Database Scenario: A stakeholder wants to know which customer segments are at risk of churning based on low feature engagement. You have: 'users' (user_id, segment, signup_date), 'feature_events' (event_id, user_id, feature_name, event_date). Write a SQL query to find segments where more than 40% of users activated fewer than 3 distinct features in their first 30 days.

ANS

UNIT 1 — Problem Framing & The Art of the Question

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A) To create visual dashboards quickly
B) To move from a surface-level symptom to the actual root cause of a problem ✓
C) To define budget constraints before analysis
D) To identify the number of stakeholders in a project

ANS

Correct Answer: B

The 5 Whys technique is a root-cause analysis method developed by Sakichi Toyoda and used in Toyota's manufacturing system. In data analysis, it acts as a mental drill — you keep asking 'Why?' until you reach an actionable root cause rather than just treating surface symptoms. The key insight is that what a stakeholder presents as the problem is often just a symptom, not the real issue.

Q2 Which framework is used to ensure a data question is well-formed and answerable?

A) GDPR Framework
B) Fishbone Diagram
C) SMART Question Framework ✓
D) Agile Sprint Retrospective

ANS

Correct Answer: C

The SMART framework stands for Specific, Measurable, Achievable (with available data), Relevant (to the business problem), and Time-bound. When converting vague business concerns into data questions, applying SMART criteria ensures the question can actually be answered with the data at hand, within a defined timeframe, and produces results that matter to the business.

Q3 What is the difference between a 'Metric' and a 'KPI'?

A) Metrics are visualizations; KPIs are raw numbers
B) A metric is any measurable number; a KPI is a metric tied to a strategic goal with a target ✓
C) KPIs are used only in healthcare; metrics are used everywhere
D) Both terms are interchangeable and mean the same thing

ANS

Correct Answer: B

A metric is simply any measurable number — for example, 'page views' or 'daily signups.' A KPI (Key Performance Indicator) is a metric that has been elevated to strategic importance with a defined target and timeline — for example, 'page views must reach 1 million per month by Q4.' Every KPI is a metric, but not every metric is a KPI.

Q4 In the Constraint Triangle model for analysis projects, which of the following is correct?

A) You can fully optimize Scope, Time, and Cost simultaneously
B) Only Time and Budget matter; Scope is flexible
C) You can optimize two but must sacrifice one ✓
D) The triangle applies only to software engineering, not data analysis

ANS

Correct Answer: C

The Constraint Triangle (also called the Iron Triangle or Triple Constraint) is a classic project management model with three vertices: Scope, Time, and Cost. The fundamental rule is that you can fully optimize any two of the three, but the third will be compromised. For example, if a stakeholder needs the full analysis scope in half the time, the cost (resources) must increase. Data analysts use this framework to negotiate realistic expectations with stakeholders.

Q5 What is a 'Guardrail Metric' in the context of KPI setting?

A) The primary metric the entire company optimizes for

B) A metric used only in manufacturing quality control

C) A secondary metric that must NOT decline while optimizing the primary goal ✓

D) A leading indicator that predicts future performance

ANS

Correct Answer: C

Guardrail metrics are protective secondary metrics defined alongside the primary KPI. They act as safety boundaries — ensuring that while you chase one goal, you don't accidentally break something else. For example, if you're optimizing for conversion rate (primary KPI), the guardrail might be 'Customer Satisfaction Score must stay above 4.2.' This prevents teams from gaming one metric at the expense of overall business health.

LEVEL 2: Interviewer Test

15 Intermediate to Semi-Advanced Questions — Descriptive, Scenario, Coding & Database

SECTION A — Descriptive Questions (4 Questions)

Q1 [Descriptive] Explain the '5 Whys' technique and how a data analyst should apply it before starting an analysis. Why is stopping at the first 'Why' dangerous?

ANS

Understanding the Concept:

So, the 5 Whys is essentially a diagnostic tool — it's like peeling back layers of an onion to get to the core issue. The technique works by asking 'Why?' repeatedly, where each answer becomes the prompt for the next question. It was originally invented by Sakichi Toyoda for Toyota's production system, but it translates perfectly into data analytics.

How I'd Apply It as an Analyst:

When a stakeholder says something vague like 'our sales are down,' I wouldn't immediately open SQL and start pulling numbers. Instead, I'd sit with them and ask:

Why 1: Why are sales down? → Fewer customers are completing purchases.

Why 2: Why are fewer customers completing purchases? → Checkout abandonment increased 35%.

Why 3: Why did abandonment increase? → Users drop off at the payment step.

Why 4: Why do they drop off at payment? → New payment gateway was introduced.

Why 5: Why does the new gateway cause drop-offs? → It added friction with 3 extra fields on mobile.

Why Stopping Early is Dangerous:

If I stopped at Why 1 — 'fewer customers are completing purchases' — I might build a dashboard tracking purchase volume and that's it. But the real, actionable insight is about the payment gateway on mobile. Stopping early means solving the symptom, not the root cause. The business would keep seeing the same problem without any real fix.

Q2 [Descriptive] What is the difference between a 'Leading Indicator' and a 'Lagging Indicator'? Give an industry example of each and explain when an analyst should prioritize one over the other.

ANS

Core Distinction:

The way I think about it — lagging indicators tell you what already happened, like reading the news. Leading indicators tell you what's about to happen, like watching the weather forecast. Both matter, but for different decisions.

Lagging Indicator Example:

Revenue for last quarter — this is the most common lagging indicator. By the time you see it, the quarter is over. You can learn from it, but you can't change it. Similarly, Employee Turnover Rate for the past year is lagging — the damage is already done.

Leading Indicator Example:

In SaaS, 'Number of features activated in the first 7 days' is a strong leading indicator of whether a user will churn within 90 days. In sales, 'Number of proposals sent this week' leads revenue next month. In HR, 'Engagement survey scores' predict resignation rates 6 months ahead.

When to Prioritize Each:

If leadership needs to evaluate past performance — use lagging indicators. If the goal is to intervene and prevent a bad outcome — leading indicators are far more valuable. The ideal KPI dashboard always includes both: lagging indicators for accountability, and leading indicators for early warning. As an analyst, I'd always ask: do we need to measure what happened, or predict what's coming?

Q3 [Descriptive] Describe the complete process of converting a vague business problem into a precise, answerable data question. Walk through each step using the SMART criteria and dimensioning approach.

ANS

Why This Step Matters:

Honestly, this is where most analysis goes wrong — not in the SQL or the visualization, but right at the start when someone asks a fuzzy question like 'Why is our app not performing well?' If I take that question literally and start pulling every metric I can find, I'll produce a 20-page report that answers nothing specific.

Step-by-Step Conversion Process:

Step 1 — Identify the Business Outcome: What is the business actually worried about? In this case, 'app not performing well' likely means user engagement is dropping or acquisition costs are rising.

Step 2 — Identify the Key Metric: What number quantifies this outcome? For engagement, I'd look at DAU/MAU ratio (Daily Active Users / Monthly Active Users). For acquisition, I'd look at CAC (Customer Acquisition Cost).

Step 3 — Add Dimensions (Who/What/When/Where): Which platform — iOS or Android? Which user segment — new vs. returning? Which time period — this quarter vs. last quarter?

Step 4 — Apply SMART: Is the question Specific (yes — DAU/MAU by platform), Measurable (yes — it's a ratio), Achievable with available data (check), Relevant to the business concern (yes), and Time-bound (this quarter vs last quarter)?

Step 5 — Confirm Answerability: Do we actually have this data? Is the event tracking in place to compute DAU/MAU by platform?

Result:

The final data question becomes: 'What is the DAU/MAU ratio by platform (iOS vs Android) in Q3 2025 compared to Q2 2025, broken down by new vs. returning users?' That's a question I can actually answer with data.

Q4 [Descriptive] Why is a Stakeholder Discovery Session important before starting any data analysis? What are the risks of skipping it, and what are the key questions an analyst must ask during such a session?

ANS

The Core Argument:

I think of the discovery session as the 'briefing before the mission.' Without it, you're essentially working blind — making assumptions about what the stakeholder wants, which metrics matter, and what a good result even looks like. And assumptions in data analysis are expensive mistakes.

Risks of Skipping It:

The biggest risk is rework — you spend two weeks building a beautiful dashboard only to be told 'that's not what I meant.' Beyond the wasted time, there's a credibility hit. Stakeholders lose trust in the analyst as a strategic partner. There's also scope creep risk — without an agreed-upon scope upfront, stakeholders keep adding requests mid-analysis.

Key Questions to Ask:

Context Questions: 'In your own words, what problem are you trying to solve?' and 'How is this problem currently impacting the business?'

Depth Questions: 'What decision will be made based on these findings?' and 'What does success look like to you?'

Constraint Questions: 'When do you need this by?' and 'Are there topics we should NOT include?'

Validation: 'So if I understood correctly, you need X by [date], and success means Y — is that right?'

The Key Output:

After the session, I always send back a written Discovery Summary for sign-off before touching any data. This becomes the contract — it prevents goalposts from moving later.

SECTION B — Scenario-Based Descriptive Questions (4 Questions)

Q1 [Scenario] Scenario: The Head of Marketing at an e-commerce company tells you: 'Our email campaigns aren't working. Can you look into it?' Walk through how you would apply the 5 Whys technique and then convert the finding into a precise data question.

ANS

My Approach:

The first thing I'd recognize is that 'email campaigns aren't working' is a symptom, not a root cause. Before I open any reporting tool, I'd sit with the Head of Marketing and run a 5 Whys chain.

5 Whys Chain:

Why 1: Why aren't email campaigns working? → Open rates are very low, around 12%.

Why 2: Why are open rates low at 12%? → Subject lines are generic and not personalized.

Why 3: Why are subject lines not personalized? → We're sending the same email to all 200,000 subscribers.

Why 4: Why are we sending to everyone? → We don't have audience segmentation in place.

Why 5: Why don't we have segmentation? → Customer purchase behavior hasn't been tagged in the CRM.

Root Cause:

The root cause is lack of CRM customer segmentation, not a creative or content problem as the stakeholder assumed.

Converted Data Question:

'What is the open rate, click-through rate, and conversion rate of weekly product launch emails for each customer purchase behavior segment (first-time buyers, repeat buyers, lapsed buyers) over the last 3 months?' — This tells us exactly which segments to target and what messaging is resonating.

Q2 [Scenario] Scenario: You are a new data analyst at a healthcare startup. The Clinical Director says: 'We need to reduce patient readmissions. Can you run an analysis?' Identify the constraints you would check before starting, and explain how you would present these constraints to the Clinical Director.

ANS

Constraint Assessment I Would Run:

Before writing a single line of SQL, I'd go through a structured constraint checklist:

Data Availability:

First, I'd ask: do we actually have readmission data — specifically, discharge dates and re-admission dates linked by patient ID? How far back does this data go — is it 6 months or 3 years? Is patient data clean and complete, or are there gaps in discharge records?

Compliance Constraints:

Healthcare is HIPAA-governed in the US. Patient data cannot be used in its raw form. I'd need to work with de-identified, tokenized patient IDs. I'd immediately check with the data governance team about approved query patterns and whether any analysis requires IRB review.

Time Constraints:

I'd ask the Clinical Director: is there a hard deadline — for example, before a board meeting or accreditation review? This determines whether I do a quick 30-day preliminary look or a full 12-month trend analysis.

How I'd Present This:

I'd share a one-page Constraint Summary saying: 'Here's what I found. Our patient data goes back 18 months and is available in anonymized form. Given compliance requirements and the

2-week deadline, I recommend we scope this to a 12-month readmission rate analysis for patients aged 65+ in the cardiology ward — does that align with your priority?' This shows I'm a strategic partner, not just a technician.

Q3 [Scenario] Scenario: A Product Manager tells you: 'Users are churning. Our retention is terrible.' Define three KPIs you would propose for this situation. For each KPI, specify the formula, target, and a guardrail metric.

ANS

My Response to the PM:

'Terrible retention' is a vague business feeling. Before we can fix it, we need to measure it precisely. Here are three KPIs I'd define:

KPI 1: Monthly Churn Rate

Formula: (Users Lost in Month / Total Users at Month Start) × 100

Target: Below 3% per month (industry benchmark for SaaS)

Guardrail: New User Acquisition Rate must not decline below 8% MoM while we optimize for retention

KPI 2: Day-30 Retention Rate

Formula: (Users still active on Day 30 / Users acquired on Day 1) × 100

Target: Above 40% (strong benchmark for consumer SaaS apps)

Guardrail: Customer Support Ticket Volume must not increase — if users are retained through forced friction, that's not real retention

KPI 3: Feature Adoption Rate (Leading Indicator)

Formula: (Users who activated 3+ core features in first 7 days / Total New Users) × 100

Target: Above 55% within first week

Guardrail: Onboarding Completion Rate must stay above 70% — if we add features but break onboarding, we lose users earlier

The reason I include the Feature Adoption Rate as a leading indicator is that low early-feature adoption almost always predicts churn — users who don't discover value quickly, leave quickly.

Q4 [Scenario] Scenario: The VP of Sales requests a '5-year revenue trend analysis' to be delivered in 3 business days. After checking your data systems, you discover the database only holds 2 years of data, and 15% of revenue records have a missing region field. How do you handle this professionally?

ANS

Constraint Reality Check:

This is actually a very common situation — the stakeholder's ask doesn't match data reality. The wrong response is to silently deliver 2 years of data and hope no one notices. The right response is to surface this immediately, professionally, and with a proposed solution.

What I Would Do:

Step 1 — Verify: I'd confirm that the database genuinely only holds 2 years (24 months) of revenue records, and that the 15% missing region field is random (not concentrated in one time period, which would bias the analysis).

Step 2 — Communicate Proactively: I'd reach out to the VP before the deadline, not after: 'I wanted to flag two data constraints I've identified. First, our revenue database holds 24 months of records, not 60. Second, 15% of records are missing the region field. I want to propose an adjusted scope.'

Step 3 — Propose a Solution: 'I recommend delivering a 2-year trend with full granularity by Day 3 as planned. For the missing region data, I'll tag those as Unattributed and note the caveat clearly in the report. If you need the 5-year view, I can check with the data warehouse team about archived records — that would add 5-7 business days.'

Why This Matters:

Analysts who surface constraints early are trusted. Analysts who deliver wrong or incomplete work without warning lose credibility permanently. The discovery session and constraint audit exist precisely to avoid this scenario.

SECTION C — Coding Questions (4 Questions)

Q1 [Coding] Write a Python function that implements the 5 Whys chain interactively. The function should accept a problem statement, prompt the user to enter answers for each 'Why', and return the root cause along with the full chain.

ANS

My Thinking:

This is a clean, practical implementation. The idea is to capture the iterative chain — each answer becomes the next prompt — and store the full reasoning path. I'd build it as an interactive CLI tool.

Python Code:

```
def five_whys(problem_statement, max_whys=5):
 """
 Runs an interactive 5 Whys root-cause analysis session.
 Returns the root cause and the full reasoning chain.
 """
 print(f"\n=== 5 WHYS ANALYSIS ===" )
 print(f"Problem Statement: {problem_statement}\n")

 chain = [] # Stores each (why, answer) pair
 current_why = problem_statement
 root_cause = None

 for i in range(1, max_whys + 1):
 print(f"Why {i}: Why is '{current_why}' happening?")
 answer = input(f" Answer {i}: ").strip()

 if not answer:
 print("No answer provided. Stopping here.")
 root_cause = current_why
 break

 chain.append({
 "why_number" : i,
 "question" : f"Why is '{current_why}' happening?",
 "answer" : answer
 })
 current_why = answer # Each answer = next Why prompt

 if i == max_whys:
 root_cause = answer
 print(f"\n☑ ROOT CAUSE IDENTIFIED: {root_cause}")

 # Build result summary
 result = {
 "problem" : problem_statement,
 "chain" : chain,
 "root_cause" : root_cause,
 "depth" : len(chain)
 }
 return result


# ── Example Usage ──
if __name__ == "__main__":
 problem = "Revenue dropped last quarter"
 analysis = five_whys(problem)
```

```
print("\n=== FULL CHAIN ===")
for step in analysis["chain"]:
print(f" Why {step['why_number']}: {step['question']}")
print(f" → {step['answer']}")
print(f"\nRoot Cause: {analysis['root_cause']}")
```

Key Design Decisions:

I used a loop rather than recursion for simplicity and readability. The chain list stores structured data (not just strings) so the result is easy to export to JSON or a report. The function is also parameterized with max_whys so you can run fewer whys for simpler problems.

Q2 [Coding] Write a Python function that takes a business problem description as input and generates a SMART data question by prompting the user for metric, dimension, and time period. Validate that all SMART criteria are met before returning.

ANS

My Approach:

The SMART criteria need to be operationalized into checkable fields. I'll capture each one separately and build the final question programmatically.

```
def build_smart_question(business_problem):
    """
    Converts a vague business problem into a SMART data question.
    Validates all SMART criteria before returning the question.
    """
    print(f"\n=== SMART QUESTION BUILDER ===")
    print(f"Business Problem: {business_problem}\n")

    smart = {}
    errors = []

    # S — Specific: What metric?
    smart["metric"] = input("S (Specific) — What metric measures this? "
    "(e.g., Churn Rate, Conversion Rate): ").strip()

    # M — Measurable: How is it calculated?
    smart["formula"] = input("M (Measurable) — How is it calculated? "
    "(e.g., Lost / Total × 100): ").strip()

    # A — Achievable: Do we have the data?
    has_data = input("A (Achievable) — Do you have this data available? "
    "(yes/no): ").strip().lower()
    smart["achievable"] = has_data == "yes"
```

```
 # R — Relevant: What dimension / segment?
 smart["dimension"] = input("R (Relevant) — Which dimension/segment? "
 "(e.g., by region, by customer type): ").strip()

 # T — Time-bound: What time period?
 smart["time_period"] = input("T (Time-bound) — What time period? "
 "(e.g., Q3 2025, last 6 months): ").strip()

 # ── Validation ──
 if not smart["metric"] : errors.append("Missing: Specific metric")
 if not smart["formula"] : errors.append("Missing: Measurable formula")
 if not smart["achievable"] : errors.append("Not Achievable: Data
unavailable")
 if not smart["dimension"] : errors.append("Missing: Relevant dimension")
 if not smart["time_period"]: errors.append("Missing: Time-bound period")

 if errors:
 print("\n❌ SMART Validation Failed:")
 for e in errors: print(f" • {e}")
 return None

 # ── Build Data Question ──
 data_question = (
 f"What is the {smart['metric']} ({smart['formula']}) "
 f"{smart['dimension']} during {smart['time_period']}?"
 )

 print(f"\n✅ SMART Data Question:\n {data_question}")
 return {"smart_components": smart, "data_question": data_question}


# ── Example Usage ──
result = build_smart_question("We are losing customers")
if result:
 print("\nGenerated:", result["data_question"])
```


Q3 [Coding] Write a Python class called 'KPITracker' that allows analysts to define KPIs, log current values, check if targets are met, and flag guardrail breaches. Include methods: add_kpi(), log_value(), check_status(), and summary_report().

ANS **Design Thinking:**

A KPI tracker needs to be structured as a dictionary of KPI records. Each KPI has a target, a guardrail, and a history of logged values. I'll make each method intuitive and return clear status strings.

```
from datetime import datetime

class KPITracker:
 """Manages KPI definitions, value logging, and target monitoring."""

 def __init__(self):
 self.kpis = {} # {kpi_name: kpi_record}

 def add_kpi(self, name, formula, target, direction="above",
 guardrail_name=None, guardrail_limit=None,
 guardrail_direction="above", time_bound=None):
 """
 Registers a new KPI.
 direction: 'above' = higher is better | 'below' = lower is better
 """
 self.kpis[name] = {
 "formula" : formula,
 "target" : target,
 "direction" : direction,
 "time_bound" : time_bound,
 "guardrail_name" : guardrail_name,
 "guardrail_limit" : guardrail_limit,
 "guardrail_direction": guardrail_direction,
 "history" : [] # list of {date, value, guardrail_value}
 }
 print(f"KPI '{name}' registered. Target: {target} ({direction}).")

 def log_value(self, name, value, guardrail_value=None):
 """Logs a new observed value for the KPI."""
 if name not in self.kpis:
 raise ValueError(f"KPI '{name}' not found. Register it first.")
 entry = {
 "date" : datetime.now().strftime("%Y-%m-%d"),
 "value" : value,
 "guardrail_value" : guardrail_value
 }
 self.kpis[name]["history"].append(entry)
 print(f"Logged {name}: {value} on {entry['date']}")

 def check_status(self, name):
 """Returns status of a KPI: ON_TRACK, OFF_TRACK, GUARDRAIL_BREACH."""
 kpi = self.kpis.get(name)
 if not kpi or not kpi["history"]:
```

```
 return "NO_DATA"

 latest = kpi["history"][-1]
 val, target = latest["value"], kpi["target"]

 # Check primary KPI
 on_track = (val >= target if kpi["direction"] == "above"
 else val <= target)

 # Check guardrail
 guardrail_breach = False
 if kpi["guardrail_limit"] and latest["guardrail_value"] is not None:
 gv, gl = latest["guardrail_value"], kpi["guardrail_limit"]
 gd = kpi["guardrail_direction"]
 guardrail_breach = (gv < gl if gd == "above" else gv > gl)

 if guardrail_breach: return "GUARDRAIL_BREACH"
 return "ON_TRACK" if on_track else "OFF_TRACK"

 def summary_report(self):
 """Prints a full status report for all tracked KPIs."""
 print("\n===== KPI SUMMARY REPORT =====")
 for name, kpi in self.kpis.items():
 status = self.check_status(name)
 latest_val = kpi["history"][-1]["value"] if kpi["history"] else "N/A"
 icon = {"ON_TRACK": "✅", "OFF_TRACK": "❌",
 "GUARDRAIL_BREACH": "⚠", "NO_DATA": "❓"}.get(status, "?")
 print(f" {icon} {name}: {latest_val} | Target: {kpi['target']} |
{status}")
 print("==============================\n")


# ── Example Usage ──
tracker = KPITracker()
tracker.add_kpi("Customer Churn Rate", "Lost/Total×100",
 target=5, direction="below",
 guardrail_name="NPS Score", guardrail_limit=40,
 guardrail_direction="above", time_bound="Q4 2025")
tracker.log_value("Customer Churn Rate", value=4.2, guardrail_value=42)
tracker.log_value("Customer Churn Rate", value=5.8, guardrail_value=38)
tracker.summary_report()
```

Q4 [Coding] Write a Python function 'constraint_audit()' that accepts analysis parameters (required_history_years, deadline_days, records_count) and evaluates feasibility. It should return a structured constraint report with recommendations.

ANS

Approach:

The constraint audit needs to programmatically check each type of constraint and produce an actionable report — the kind you'd share with a stakeholder before starting work.

```
def constraint_audit(required_history_years, available_history_years,
 deadline_days, estimated_analysis_days,
 records_count, missing_pct,
 has_pii=False, has_budget=True):
 """
 Evaluates analysis feasibility across all constraint dimensions.
 Returns a structured constraint report with recommendations.
 """
 report = {
 "constraints" : [],
 "risks" : [],
 "recommendations": [],
 "feasible" : True
 }

 # ── 1. Data Availability ──
 if available_history_years < required_history_years:
 gap = required_history_years - available_history_years
 report["constraints"].append(
 f"DATA: Only {available_history_years}yr available; "
 f"{required_history_years}yr requested (gap: {gap}yr)"
 )
 report["recommendations"].append(
 f"Reduce scope to {available_history_years}-year analysis, "
 f"or check data warehouse for archived records."
 )
 report["feasible"] = False

 # ── 2. Data Quality ──
 if missing_pct > 10:
 report["risks"].append(
 f"QUALITY: {missing_pct}% of records have missing fields — "
 f"analysis may be biased."
 )
 report["recommendations"].append(
 "Filter to clean records only and add caveat to report. "
 "Investigate if missing data is random or systematic."
 )

 # ── 3. Time Feasibility ──
 if estimated_analysis_days > deadline_days:
 report["constraints"].append(
```

```
f"TIME: Analysis needs {estimated_analysis_days} days; "
f"deadline is {deadline_days} days."
)
report["recommendations"].append(
"Negotiate: deliver preliminary insight by deadline, "
"full analysis in extended timeline."
)
report["feasible"] = False

# ── 4. Scale / Performance ──
if records_count > 10_000_000:
report["risks"].append(
f"SCALE: {records_count:,} records may require cloud compute. "
f"Standard tools may timeout."
)
report["recommendations"].append(
"Use stratified sampling (20-30%) for exploratory analysis "
"before running full dataset."
)

# ── 5. Compliance ──
if has_pii:
report["constraints"].append(
"COMPLIANCE: Data contains PII — GDPR/HIPAA review required."
)
report["recommendations"].append(
"Work with anonymized/tokenized IDs only. "
"Request governance approval before proceeding."
)

# ── 6. Budget ──
if not has_budget:
report["risks"].append(
"BUDGET: No compute budget allocated for large-scale processing."
)

# ── Print Report ──
print("\n===== CONSTRAINT AUDIT REPORT =====")
print(f"Feasibility: {'✅ FEASIBLE' if report['feasible'] else '❌ NEEDS ADJUSTMENT'}")
print("\nConstraints Identified:")
for c in report["constraints"]: print(f" ⛔ {c}")
print("\nRisks Flagged:")
for r in report["risks"]: print(f" ⚠️ {r}")
print("\nRecommendations:")
for i, rec in enumerate(report["recommendations"], 1):
print(f" {i}. {rec}")
print("==================================\n")
```

```
 return report


# ── Example: VP of Sales 5-year analysis request ──
constraint_audit(
 required_history_years = 5,
 available_history_years = 2,
 deadline_days = 3,
 estimated_analysis_days= 7,
 records_count = 15_000_000,
 missing_pct = 15,
 has_pii = True,
 has_budget = False
)
```

SECTION D — Database Scenario-Based Questions (3 Questions)

Q1 [DB Coding] Database Scenario: You have a table 'orders' with columns: order_id, customer_id, order_date, total_amount, product_category, region, status. Write a SQL query to identify the top 3 product categories by return rate (status = 'Returned') for the last 6 months, along with total returns and total orders per category.

ANS

Understanding the Business Need:

This is a direct application of Topic 1.2 — converting a business problem ('returns are rising') into a data question ('Which product categories have the highest return rate in the last 6 months?'). The SQL needs to aggregate by category, compute both counts, and calculate the rate.

SQL Query:

```
-- Top 3 Product Categories by Return Rate (Last 6 Months)
SELECT
 product_category,
 COUNT(*) AS total_orders,
 SUM(CASE WHEN status = 'Returned' THEN 1 ELSE 0 END)
 AS total_returns,
 ROUND(
 SUM(CASE WHEN status = 'Returned' THEN 1 ELSE 0 END) * 100.0
 / COUNT(*),
 2
 ) AS return_rate_pct
FROM
 orders
```

```
WHERE
 order_date >= CURRENT_DATE - INTERVAL '6 months' -- Last 6 months filter
 AND order_date < CURRENT_DATE
GROUP BY
 product_category
HAVING
 COUNT(*) >= 50 -- Exclude categories with too few orders (statistical
noise)
ORDER BY
 return_rate_pct DESC
LIMIT 3;
```

Query Design Handbook:

The HAVING COUNT(*) >= 50 guard is important — without it, a category with 2 orders and 1 return would show as 50% return rate, which is statistically meaningless. The ROUND(..., 2) keeps percentages clean for reporting. I'd also add a CTE version if the stakeholder wants to filter by region as well.

Q2 [DB Coding] Database Scenario: You have two tables — 'customers' (customer_id, signup_date, plan_type, region) and 'subscriptions' (sub_id, customer_id, start_date, end_date, status). Write a SQL query to calculate the monthly churn rate for the last 12 months, broken down by plan_type. Define 'churned' as status = 'cancelled'.

ANS

KPI in SQL:

This query operationalizes the Churn Rate KPI we defined in Topic 1.3. The challenge here is computing 'customers at start of month' vs 'customers lost during month' — which requires a time-series calculation.

SQL Query:

```
-- Monthly Churn Rate by Plan Type — Last 12 Months
WITH monthly_base AS (
 -- Count active customers at the start of each month per plan_type
 SELECT
 DATE_TRUNC('month', s.start_date) AS cohort_month,
 c.plan_type,
 COUNT(DISTINCT s.customer_id) AS active_at_start
 FROM subscriptions s
 JOIN customers c ON s.customer_id = c.customer_id
 WHERE
 s.start_date >= CURRENT_DATE - INTERVAL '12 months'
 AND s.status IN ('active', 'cancelled') -- Include both
 GROUP BY 1, 2
),
```

```
monthly_churned AS (
 -- Count customers who cancelled within each month
 SELECT
 DATE_TRUNC('month', s.end_date) AS churn_month,
 c.plan_type,
 COUNT(DISTINCT s.customer_id) AS churned_count
 FROM subscriptions s
 JOIN customers c ON s.customer_id = c.customer_id
 WHERE
 s.status = 'cancelled'
 AND s.end_date >= CURRENT_DATE - INTERVAL '12 months'
 GROUP BY 1, 2
)
SELECT
 b.cohort_month,
 b.plan_type,
 b.active_at_start,
 COALESCE(ch.churned_count, 0) AS churned,
 ROUND(
 COALESCE(ch.churned_count, 0) * 100.0
 / NULLIF(b.active_at_start, 0),
 2
 ) AS churn_rate_pct
FROM monthly_base b
LEFT JOIN monthly_churned ch
 ON b.cohort_month = ch.churn_month
 AND b.plan_type = ch.plan_type
ORDER BY
 b.cohort_month DESC,
 churn_rate_pct DESC;
```

Design Decisions:

I used CTEs to separate the logic cleanly — one CTE for active counts, one for churned counts. NULLIF(active_at_start, 0) prevents division-by-zero errors. COALESCE(churned_count, 0) handles months where no churn occurred for a plan type. This is production-grade SQL that a BI team would actually deploy.

Q3 [DB Coding] Database Scenario: A stakeholder wants to know which customer segments are at risk of churning based on low feature engagement. You have: 'users' (user_id, segment, signup_date), 'feature_events' (event_id, user_id, feature_name, event_date). Write a SQL query to find segments where more than 40% of users activated fewer than 3 distinct features in their first 30 days.

ANS **Leading Indicator Query:**

This query computes the Feature Adoption Rate leading indicator from our KPI discussion. Users who activate fewer than 3 features in 30 days are high-churn risk. We need to first compute per-user adoption, then aggregate by segment to find at-risk segments.

SQL Query:

```
-- At-Risk Segments: <3 Features Activated in First 30 Days
WITH user_feature_adoption AS (
 -- Count distinct features each user activated in their first 30 days
 SELECT
 u.user_id,
 u.segment,
 u.signup_date,
 COUNT(DISTINCT fe.feature_name) AS distinct_features_used
 FROM users u
 LEFT JOIN feature_events fe
 ON fe.user_id = u.user_id
 AND fe.event_date >= u.signup_date
 AND fe.event_date < u.signup_date + INTERVAL '30 days'
 GROUP BY
 u.user_id, u.segment, u.signup_date
),
segment_risk_summary AS (
 -- Aggregate by segment: total users, low-adoption users, rate
 SELECT
 segment,
 COUNT(*) AS total_users,
 SUM(CASE WHEN distinct_features_used < 3 THEN 1 ELSE 0 END)
 AS low_adoption_users,
 ROUND(
 SUM(CASE WHEN distinct_features_used < 3 THEN 1 ELSE 0 END)
 * 100.0 / NULLIF(COUNT(*), 0),
 1
 ) AS low_adoption_pct
 FROM user_feature_adoption
 GROUP BY segment
 HAVING COUNT(*) >= 20 -- Minimum segment size for reliability
)
SELECT
 segment,
 total_users,
 low_adoption_users,
 low_adoption_pct,
 CASE
 WHEN low_adoption_pct > 60 THEN 'HIGH RISK'
 WHEN low_adoption_pct > 40 THEN 'AT RISK'
 ELSE 'HEALTHY'
 END AS risk_level
FROM segment_risk_summary
```

```
WHERE
 low_adoption_pct > 40 -- Only show at-risk and high-risk segments
ORDER BY
 low_adoption_pct DESC;
```

Business Value:

The CASE statement adds a risk tier (HIGH RISK / AT RISK) which makes the output immediately actionable — the Product or Customer Success team knows exactly which segments to intervene on. The LEFT JOIN in the first CTE is critical: it ensures users with zero feature events still appear in the results with distinct_features_used = 0, rather than being silently excluded.

Unit 1 — Quick Revision Summary

Problem Framing & The Art of the Question

1.1 5 Whys Technique

Ask "Why?" repeatedly (up to 5 times) to reach the root cause of a business problem, not just the surface symptom.

1.2 Business → Data Questions

Translate vague business concerns into specific, measurable data questions using SMART criteria and dimensions.

1.3 Success Metrics / KPIs

Define KPIs (with targets and timelines) before analysis begins so everyone knows what "success" looks like.

1.4 Stakeholder Interviewing

Run structured discovery sessions to understand real needs, constraints, and expected outcomes from stakeholders.

1.5 Identifying Constraints

Audit data availability, time limits, and budget before committing to analysis scope — to set realistic expectations.

Remember: Frame the right question before you touch the data.